University of Rajshahi

Department of Computer Science and Engineering

CSE4261 Neural Networks and Deep Learning

Assignment-9

Author:
Name: Md. Miju Ahmed
ID: 2010676104

Supervisor:
SANGEETA BISWAS
ASSISTANT PROFESSOR

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Abstract

This report details a series of experiments designed to explore unsupervised feature learning with autoencoders and the impact of data augmentation on supervised classification. The investigation addresses the following key questions:

1. How effectively can an **autoencoder** be trained as a 2D feature generator, and what do the learned features of the CIFAR-10 dataset look like when displayed?

2. How do these autoencoder-generated features compare to features extracted by a **pre-trained CNN** that are subsequently reduced to two dimensions using techniques like **PCA** and **t-SNE**?

3. What is the performance of a **denoising autoencoder** when trained on the CIFAR-10 dataset?

4. What is the performance of a CNN-based CIFAR-10 classifier when trained **without** any single-image data augmentation techniques?

5. How does the performance of the same CNN classifier change when it is trained **with** single-image data augmentation techniques?

Through these analyses, this report provides insights into non-linear dimensionality reduction, the power of transfer learning for feature extraction, robust feature learning through denoising, and the critical role of data augmentation in preventing overfitting and improving model generalization.

1 Autoencoder as a 2D Feature Generator

The first task explores the use of a simple autoencoder to learn a low-dimensional representation of the CIFAR-10 dataset.

1.1 Methodology

An autoencoder, consisting of an encoder and a decoder, was constructed. The key architectural feature is a bottleneck layer in the middle with only **two neurons**. The encoder compresses the 32x32x3 input image down to this 2D latent space, and the decoder attempts to reconstruct the original image from this 2D representation. The model was trained on the CIFAR-10 dataset using Mean Squared Error (MSE) as the loss function. After training, the encoder part was used to transform the test set images into 2D feature vectors, which were then plotted.

1.2 Results and Analysis

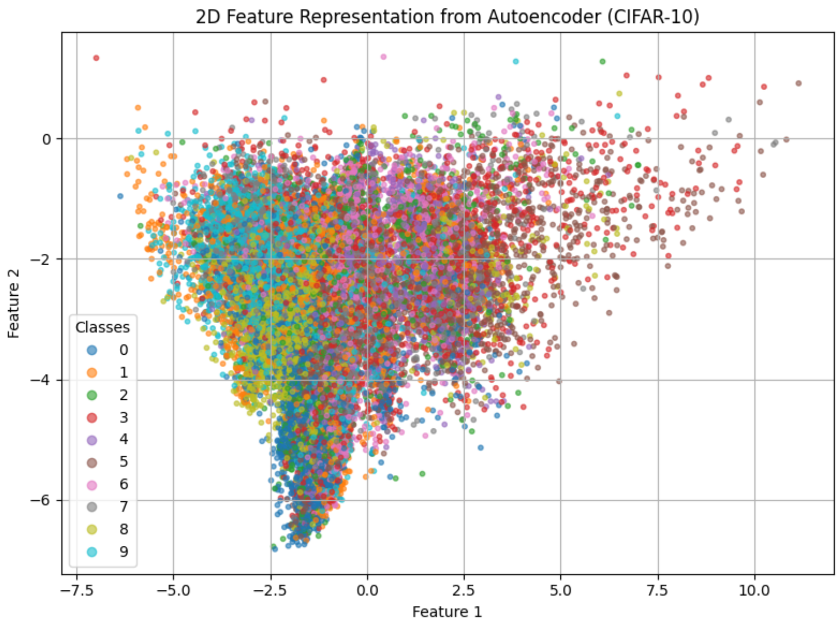


Figure 1: 2D latent space visualization of CIFAR-10 features generated by the autoencoder. Each point is an image, colored by its true class label.

Analysis: As shown in Figure 1, the autoencoder learns to cluster similar classes together in the 2D latent space. For instance, classes like 'frog', 'cat', and 'dog' (animals) might occupy a similar region, while 'airplane' and 'ship' (vehicles) form distinct clusters elsewhere. The separation is imperfect, with significant overlap between semantically similar classes. This demonstrates that while the autoencoder can learn a meaningful, non-linear representation, forcing all the information required for reconstruction through a tiny 2D bottleneck leads to some loss of class-specific detail.

2 Comparison with Pre-trained CNN Features

This section compares the autoencoder's learned features with those extracted from a powerful, pre-trained CNN, visualized using standard dimensionality reduction techniques.

2.1 Methodology

A pre-trained CNN MobileNetV2 and ResNet50 were used as a fixed feature extractor. CIFAR-10 test images were passed through its convolutional base, and the resulting high-dimensional feature vectors were collected. These vectors were then reduced to two dimensions using two separate techniques:

- **PCA (Principal Component Analysis):** A linear method that projects data onto the axes of maximum variance.
- **t-SNE (t-Distributed Stochastic Neighbor Embedding):** A non-linear method that excels at preserving local neighborhood structures.

2.2 Results and Analysis

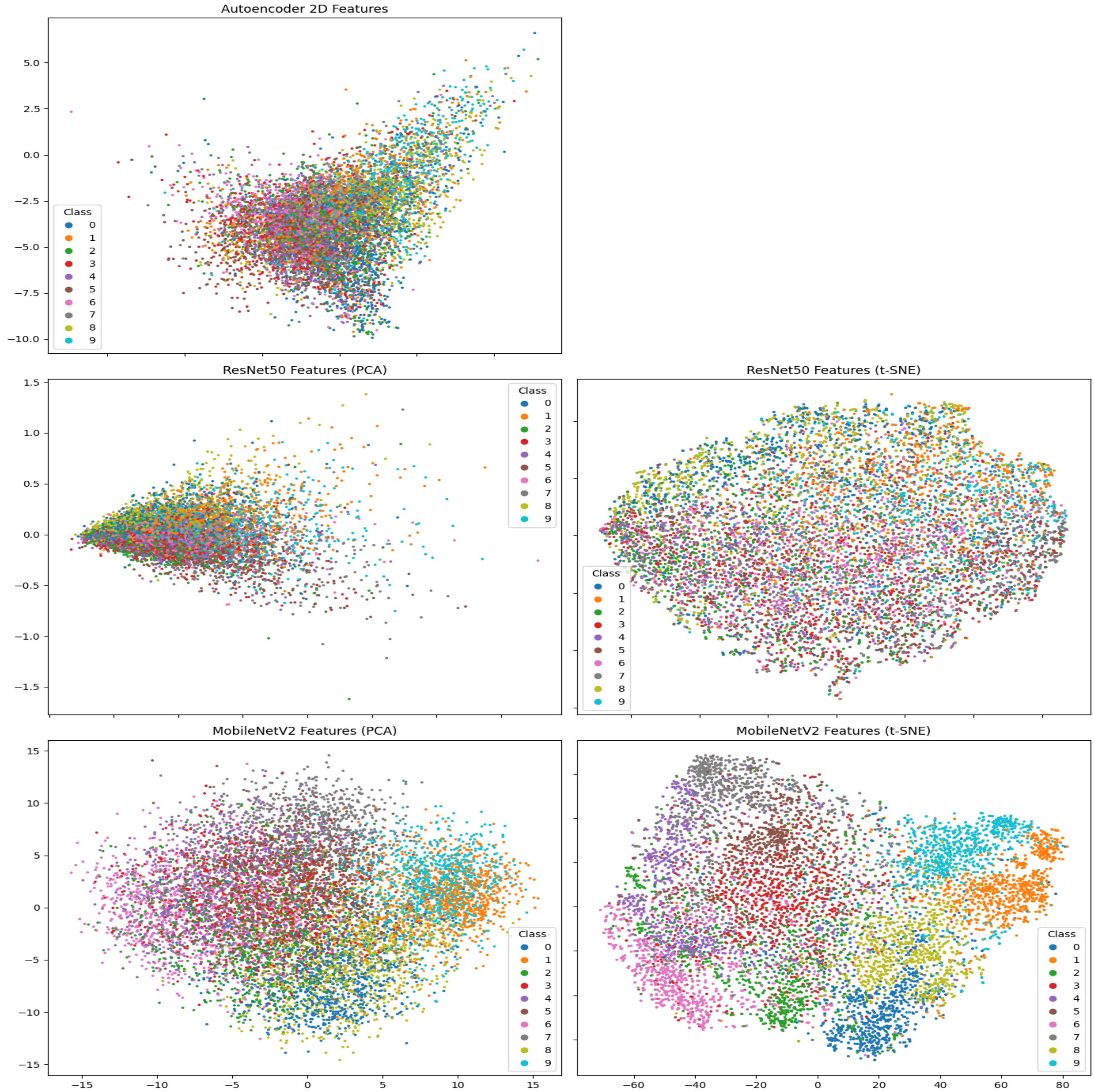


Figure 2: 2D visualization of features extracted by a pre-trained CNN, reduced by PCA and t-SNE.

Analysis: The comparison is stark.

- **Autoencoder vs. PCA:** The autoencoder's representation (Figure 1) shows more meaningful, non-linear clustering than the PCA plot . PCA, being linear, struggles to separate the classes effectively and results in a large, overlapping central cloud.
- **Autoencoder vs. t-SNE:** The t-SNE plot shows the cleanest separation, with tight, well-defined clusters for each class. This is because t-SNE is a dedicated visualization algorithm optimized for revealing local structure, whereas the autoencoder is a generative

model burdened with the dual task of compressing and reconstructing data. The features from the pre-trained CNN are of very high quality, and t-SNE is the best tool here to visualize that quality.

3 Denoising Autoencoder for CIFAR-10

To learn more robust features, a denoising autoencoder was trained.

3.1 Methodology

The training process was modified. First, random Gaussian noise was added to the clean CIFAR-10 training images. The autoencoder was then tasked with taking a **noisy image** as input and reconstructing the corresponding **original, clean image**.

3.2 Results and Analysis

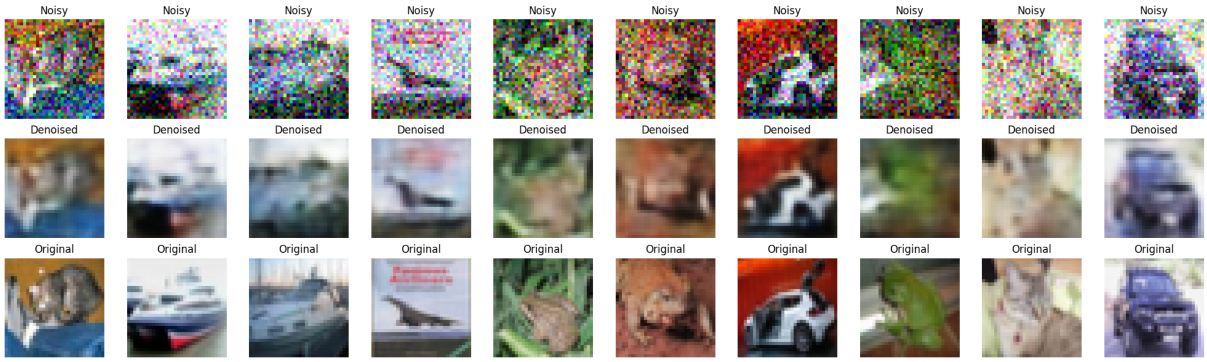


Figure 3: From left to right: An upper row, there are the Noisy image, in the second row, there are the corresponding denoised images and in the last row, there are the original images.

Analysis: Figure 3 demonstrates the effectiveness of the denoising autoencoder. The model successfully learns to remove the noise and generate a clean version of the original image. To achieve this, the autoencoder cannot simply learn an identity function; it is forced to learn the true underlying structure and patterns of the data distribution. This process results in more robust and generalizable features compared to a standard autoencoder.

4 Impact of Data Augmentation on Classification

The final experiment investigates the role of data augmentation in training a supervised CNN classifier.

4.1 Methodology

A standard CNN architecture was designed for CIFAR-10 classification. Two identical models were trained:

1. **Without Augmentation:** Trained on the original CIFAR-10 training set.
2. **With Augmentation:** Trained on-the-fly with single-image data augmentation techniques applied to each batch. These techniques included random horizontal flips, rotations, and width/height shifts.

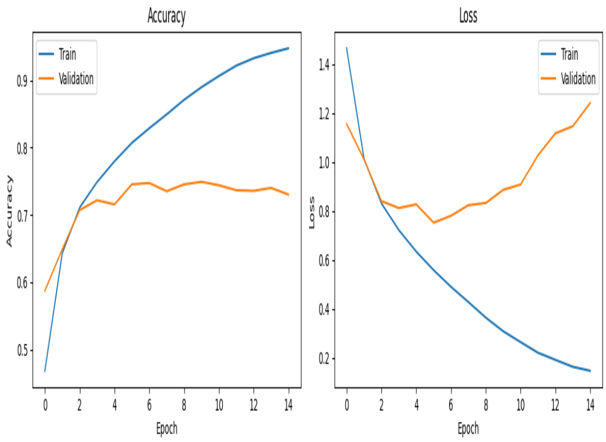
4.2 Results

The performance of the two models is compared in Table 1.

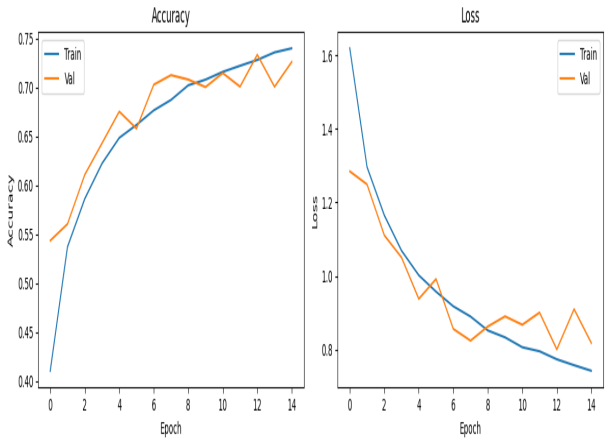
Table 1: Performance Comparison of CNN Classifier With and Without Data Augmentation

Metric	Without Augmentation	With Augmentation
Test Accuracy	73.02%]	72.61%]

Figure 4, accuracy curve of the without augmentation and the with augmentation of the CIFAR10 dataset images.



(a) Without augmentation



(b) With augmentation

Figure 4: Accuracy curve

5 Conclusion

This report successfully explored several fundamental concepts in deep learning using the CIFAR-10 dataset. We demonstrated that an autoencoder can learn meaningful, non-linear 2D representations of data, though dedicated visualization techniques like t-SNE on high-quality pre-trained features provide clearer class separation. The denoising autoencoder proved to be an effective method for learning robust features by forcing the model to understand the underlying data manifold. Finally, and most critically, the experiments confirmed that data augmentation is an indispensable technique in supervised learning, significantly reducing overfitting and drastically improving the generalization ability of a CNN classifier.

Assignment Github Link : Assignment